

BME221 DESIGN PROJECT WORK LOG – TEAM 2G

Finger prosthetic that is moved up & down with string(s) attached to a pulley (stepper motor). The finger will have a pressure sensor at the tip

Task Tracking Table (on going)

Task	Owner(s)	Date Assigned	Target Completion	Status	Notes / Dependencies
Revise and reprint finger & test bed	Guillermo Zoey	03/02	03/20	Done	
Change stepper motor to servo, and/or change pressure sensor to distance sensor	Petra	04/01	04/07	Done	
Final video and demo	All	04/05	04/16	In Progress	
Rough design + measurements of finger and test bed on solidworks	Guillermo Zoey	02/15	02/25	Done	Thicker string used to keep finger straight, while the thinner string is looped around pulley to bend the finger
Get strings & pressure sensor	Georgina	02/15	02/18	Done	Need 2 types of strings: thin nylon string, and thicker one (any material)
Simulate rotation using motor with string attached	Petra	02/15	03/04	Done	

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Print finger and test bed	Georgina Petra	02/18	03/04	Done	
Collect measurements of stepper motor, the 2 strings, etc	Georgina	02/15	02/20	Done	Measure using caliper; Measurements given to person in charge of finger and test bed design
Pressure sensor and LED code	Petra	02/15	02/19	Done	
Sketch of solidworks design and start solidworks	Guillermo Zoey	02/18	02/18	Done	Sketch the exact parts that we want to design on solidworks
“Proof of Progress” video	Georgina	02/28	03/06	Done	
Speed up motor rotation & fix wire connections	Petra	03/16	03/16	Done	Solder wires

Project Snapshot (Week 12)

Week of: 12

Major Goal for This Week: change arduino components (stepper motor -> servo motor; pressure sensor -> distance sensor), assemble all parts, working on video and demo script and editing

Technical Progress Summary

Parts have been printed, some arduino components will be changed

Questions for Instructor / TA

Issues that need escalation

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Team Process Check-In (Weekly)

- What Went Well This Week
 - We were able to print the finger and test bed
- Challenges We Faced
 - We are still trying to improve the finger design. Using just a paracord as the backbone on the finger wouldn't be stable enough, so we might need to design a joint
- What We Will Change for Next Week
 - We've been having problems with the pressure sensor not detecting any touch/pressure since it requires constant contact at a specific area. So we were thinking of changing the pressure sensor to an ultrasonic distance sensor. With the distance sensor, we can set the LED to gradually brighten as the finger moves closer to the distance sensor. This option might work better with a stepper motor. Another option is to set the LED to light up when the finger reached a certain distance threshold from the sensor, which might work better with a servo motor

Next Steps / Action Plan (*Update this section during/after meeting*)

- Key tasks for next week
- Responsible team members
- Required resources or instructor input
- Risks or uncertainties to monitor

Project Snapshot (Week 11)

Week of: 11

Major Goal for This Week: Complete revision for finger and motor test bed designs and submit for printing. Prepare for demos (in two weeks).

Technical Progress Summary

Finger Prototype: Try redesigning the original prototype to include the metal plate. If not possible, a new design will be submitted (already completed, awaiting printing).

An updated mount was sent for printing.

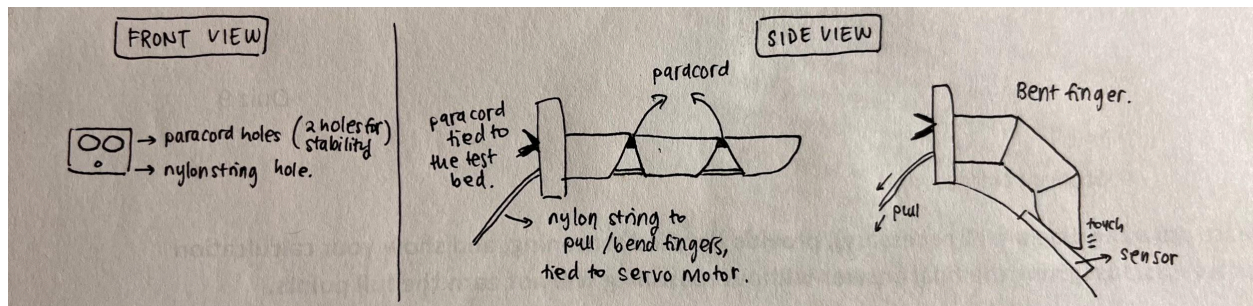
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Questions for Instructor / TA

Issues that need escalation

Team Process Check-In (Weekly)

- What Went Well This Week
 - We were able to revise design for finger, test bed, and sensor
- Challenges We Faced
 - Getting designs in on time to avoid long printing queues
 - The stepper motor is still not strong enough to bend the fingers.
- What We Will Change for Next Week
 - Get paracord
 - Start recording demos
 - We might need to switch to servo motor because the stepper motor we use is not fast/strong enough
 - We might want to change the design of the finger by using a paracord instead of a thin plastic / metal. Or we can stick to using a thin plastic / metal plate, but still change the shape of each finger component to a trapezium



Note: Need to make joint to make sure each finger part won't slip or overlap each other.

Next Steps / Action Plan *(Update this section during/after meeting)*

- Key tasks for next week
- Responsible team members
- Required resources or instructor input
- Risks or uncertainties to monitor

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- Make motor piece insert a slot instead of trying to fit it perfectly into the testbed
 - Using a drill press or wood screw to screw in motor piece
 - Make locating dimples instead of actual holes in the design
- Redesign finger: make the first digit a square for more stability
- Simplify testbed design to be less wasteful with material
- Make sensor test bed as small as possible
 - Use wedges to adjust the height
- Responsible team members
 - Guillermo & Zoey: finger and testbed revision
 - Georgina: pressure sensor testbed design
 - Petra: motor mechanics
- Required resources or instructor input
 - Drill press or wood screw
 - Thinner string
- Risks or uncertainties to monitor
 - Ensuring proper bending of the finger prototype with new design
 - Simplifying sensor testbed, taking into account friction & resolution (if using wedges)

Project Snapshot (Week 9)

Week of: 9

Major Goal for This Week: Fix test bed and finger design, make motor rotation faster, strengthen wire connections

Technical Progress Summary

We are revising our test bed & finger design and were able to fix the arduino components

Questions for Instructor / TA

Issues that need escalation

Team Process Check-In (Weekly)

- What Went Well This Week:

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- The motor is now able to rotate faster
 - We fixed the connections to the sensor by soldering the wires
- Challenges We Faced
 - The wires were loosely connected to the sensor, so the readings were inconsistent. We had to rely on tape at first.
 - The serial monitor will only print the readings of the sensor after the motor rotates clockwise and counterclockwise. However, we wanted it to print the recordings as soon as force was detected by the sensor. Applying Interrupt Service Routine (ISR) might work.
 - The batteries died pretty quickly, so it was difficult to keep it running for long during trials. Besides that, the motor specifically needs 5V battery and we're currently using 9V. Not sure if this impacts the motor performance at all.
 - The motor can now rotate faster, but it might still not be strong/fast enough to pull the fingers. We will be testing the new finger design using a sheet of metal to connect the finger parts, so hopefully it will be more elastic and easier to pull.
- What We Will Change for Next Week

Next Steps / Action Plan (*Update this section during/after meeting*)

- Key tasks for next week
- Responsible team members
- Required resources or instructor input
- Risks or uncertainties to monitor

Project Snapshot (Week 8)

Week of: 8

Major Goal for This Week: Revise finger and test bed, complete “Proof of Progress” Video

Technical Progress Summary

Parts printed but needs revision

Questions for Instructor / TA

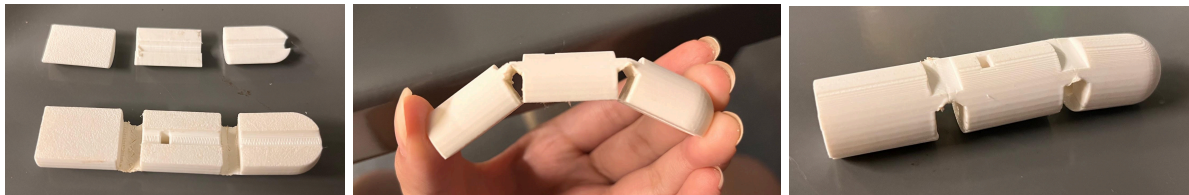
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- Would TPU be a good alternative material to use to print the finger, or is it too flimsy?
- How would you recommend securing the moving finger on a static test bed? Or should we stick to our idea (in the “What We Will Change for Next Week” section)?

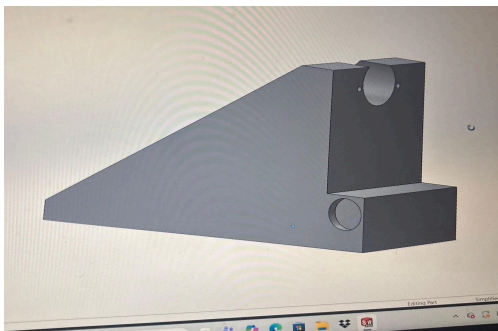
Issues that need escalation

Team Process Check-In (Weekly)

- What Went Well This Week
 - Finalized the rough design on the finger and test bed to mount the motor
 - Printed all parts (still needs further revision)
- Challenges We Faced
 - Trying to figure out how to bend the flat part that connects the finger. We initially printed 2.5mm of that flat part and had to file the middle of it so it bends. We’re concerned that it will brake if it bends too much/too often, or the motor won’t be strong enough to bend the finger
 - The flat part is not long enough, so the finger can only bend partially



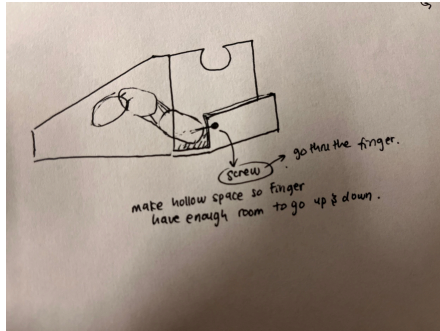
- With our current test bed design, the finger will be static. So we’re still trying to figure out the best way to stick it in the test bed while being able to bend up and down



- What We Will Change for Next Week

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- Revise design of finger: longer flat part with around 1.5–2mm thickness. Still need to file the middle so it can bend. Also make a small hole on the side to screw to the test bed
- Revise design for test bed to insert the finger: make a hole and insert a screw that goes through the finger (maybe. need suggestions)



Next Steps / Action Plan (*Update this section during/after meeting*)

- Key tasks for next week
 - Test motor capabilities to bend finger up and down
 - Proof of concept video
- Responsible team members
- Required resources or instructor input
- Risks or uncertainties to monitor

Thin metal/stainless steel so more flexible to bend, find at machine shop

Redesign the finger to have a slot in the middle of the finger to slide the steel in. can also make the top flat, so we can screw the steel on top

Project Snapshot (Week 7)

Week of: 7

Major Goal for This Week: Print parts, get arduino working

Technical Progress Summary

Code and arduino setup works well, will finalize parts design and get them printed

Questions for Instructor / TA

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Issues that need escalation

Team Process Check-In (Weekly)

- What Went Well This Week:
 - We were able to get the code running to activate LED, sensor, and motor
- Challenges We Faced:
 - Finding the right resistor for the pressure sensor (we initially used 470R). The sensor needs to be pressed hard for it to detect force and light up the LED. Experimented using 100K, 330K, 680K, 910K, and 1.5M resistors. It was more sensitive to touch now that we used the 1.5M one. Most of the time, it still needs to be “activated” by pressing on the cable connections. The touch also must stay for approx 1-4 sec and use a slight force for it to be able to light up the LED
 - Code will only print pressure/force level after printing ‘clockwise’ and ‘counter-clockwise’, even if the LED lights up when detecting force. So usually, it will print ‘force level = 0.00’ bcs it prints only after rotation, not every time it detects force. This doesn’t matter now, but might be important when we eventually test various forces.
 - Difficulty designing the holes to connect finger
- What We Will Change for Next Week:
 - Maybe make the sensor a little bit further up than the finger, so the rotation of the motor can “push” the finger to the sensor, rather than just touching it. Another option is to use weights/screws to make the finger tip heavier, adding force

Next Steps / Action Plan (*Update this section during/after meeting*)

- Key tasks for next week:
 - Test the prototype
 - Get the parts revised if needed
- Responsible team members
- Required resources or instructor input
- Risks or uncertainties to monitor

Project Snapshot (Week 6)

Week of: 6

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Major Goal for This Week: Get all materials, finalize code, design draft for finger and test bed

Technical Progress Summary

Prototypes built or tested, Data collected, Models or simulations run, Research completed. Make sure to include who contributed to the progress

Requirement	How to Measure	Status	Adjustment Plan (If failed)
Pressure sensor must detect a minimum force of _____	Check if LED feedback is detected and serial monitor shows pressure change dg Check if the LED lights up when the finger puts pressure on the sensor.		Adjust values
Finger must achieve a 90 degree bend when pulled and 0 when released	Use protractor to check contractions		Increase motor pulley diameter

Questions for Instructor / TA

Would the motor be strong enough to pull the string and bend the finger, theoretically?

Would it be better to put the sensor on the tip of the finger or on the ground for the finger to touch?

Issues that need escalation

Meeting Notes

- Split design and make channel in the middle and glue back together
 - Avoids overhang
- Make test bed to see how thick it has to be → test flexibility
- Alignment pins and/or glue
- Two pressure sensors: 1 on finger + 1 on ground
 - Have them touch each other

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- Minimize bending by making hole where plastic part is shorter

Team Process Check-In (Weekly)

- What Went Well This Week: We finalized the prototype sketches and collected all measurements to start working on solidworks.
- Challenges We Faced: Figuring out the most effective mechanical placement for the pulley/string system for proper finger flexion.
- What We Will Change for Next Week: Transition to solidworks modeling

Next Steps / Action Plan (*Update this section during/after meeting*)

- Key tasks for next week
- Responsible team members
- Required resources or instructor input
- Risks or uncertainties to monitor

Project Snapshot (Week 5)

Week of: 5

Major Goal for This Week: Finalize draft presentation & group

Technical Progress Summary

Prototypes built or tested, Data collected, Models or simulations run, Research completed. Make sure to include who contributed to the progress

Questions for Instructor / TA

Issues that need escalation

Team Process Check-In (Weekly)

- What Went Well This Week
- Challenges We Faced
- What We Will Change for Next Week

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Next Steps / Action Plan (*Update this section during/after meeting*)

- Key tasks for next week
- Responsible team members
- Required resources or instructor input
- Risks or uncertainties to monitor

Goals for video assignment:

Testbed

- file/cut down motor insert section
- Mount screw/tape motor system onto testbed

Finger Piece

- Hold down or tape finger in place

Video

- Take video of all working parts
- Create transcript
- Put video clips together